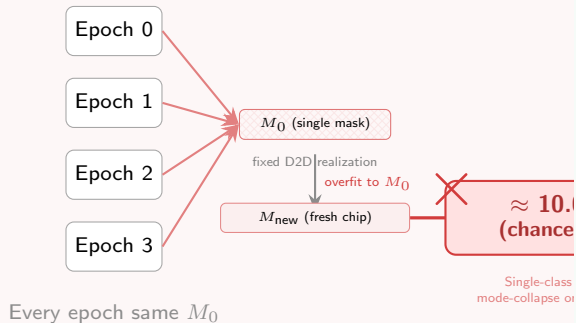


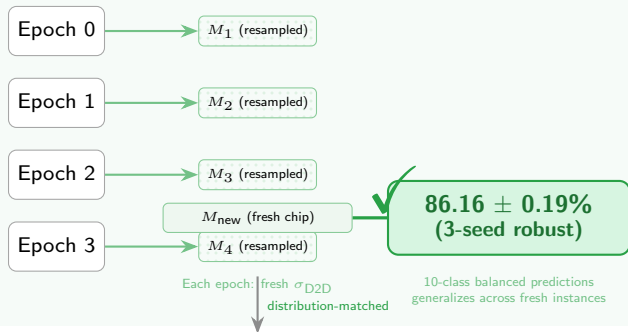
Key Mechanism: Standard HAT trains against *one* fixed hardware realization M_0 , learning to exploit its specific noise pattern. Ensemble HAT resamples $M_i \sim P(\sigma_{D2D})$ each epoch, forcing weights to learn *distribution-level* invariance rather than instance-specific compensation.

(a) Standard HAT: Single Fixed Mask



Hardware-instance overfitting

(b) Ensemble HAT: Per-Epoch Resampling



Distribution-matched training